

Week 2 Lab portfolio

Task 1

Before I could do this task, I had to install MongoDB and import the database into it.

Then I had to do some queries (The queries and results will be shown together).

\$match

Open docs

```
1 {
2   education: "Bachelor",
3   age: { $gte: 21 }
4 }
```

STAGE OUTPUT

Sample of 10 documents

_id: ObjectId('6ac79e9aef6af62527a8b2')

firstName: "Grace"

lastName: "Nelson"

gender: "Female"

age: 21

email: "g.nelson@randatmail.com"

education: "Bachelor"

salary: 5347

maritalstatus: "Single"

_id: ObjectId('6ac79e9aef6af62527a8b8')

firstName: "Charlie"

lastName: "Perkins"

gender: "Male"

age: 28

email: "c.perkins@randatmail.com"

education: "Bachelor"

salary: 3586

maritalstatus: "Single"

_id: ObjectId('6ac79e9aef6af62527a8c4')

firstName: "Helen"

lastName: "Spencer"

gender: "Female"

age: 26

email: "h.spencer@randatmail.com"

education: "Bachelor"

salary: 5987

maritalstatus: "Married"

_id: ObjectId('6ac79e9aef6af62527a8c9')

firstName: "Nicholas"

lastName: "Barrett"

gender: "Male"

age: 24

email: "n.barrett@randatmail.com"

education: "Bachelor"

salary: 5577

maritalstatus: "Married"

_id: ObjectId('6ac79e9aef6af62527a8cf')

firstName: "Helen"

lastName: "Ferguson"

gender: "Female"

age: 24

email: "r.ferguson@randatmail.com"

education: "Bachelor"

salary: 6921

maritalstatus: "Single"

_id: ObjectId('6ac79e9aef6af62527a8db')

firstName: "Aida"

lastName: "Wells"

gender: "Female"

age: 21

email: "a.wells@randatmail.com"

education: "Bachelor"

salary: 3636

maritalstatus: "Single"

Stage 1

\$group

```
1 {
2   _id: "$gender",
3   Avg: { $avg: "$age" },
4   minAge: { $min: "$age" },
5   maxAge: { $max: "$age" }
6 }
```

Output preview after \$group stage (Sample of 2 documents)

_id: "Female"

Avg: 24.064814814814813

_id: "Male"

Avg: 24.391304347826086

Stage 1

\$group

```
1 {
2   _id: "$gender",
3   Avg: { $avg: "$age" },
4   minAge: { $min: "$age" },
5   maxAge: { $max: "$age" }
6 }
```

Output preview after \$group stage (Sample of 2 documents)

_id: "Female"

Avg: 24.064814814814813

minAge: 18

maxAge: 30

_id: "Male"

Avg: 24.391304347826086

minAge: 18

maxAge: 30

Stage 1 ☒

```

1 {
2   _id: "$gender",
3   MaxSalary: {$max: "$salary"},
4   MinSalary: {$min: "$salary"},
5   AvgSalary: {$avg: "$salary"}
6 }

```

Output preview after [\\$group](#) stage (Sample of 2 documents)

_id: "Male"
MaxSalary: 9989
MinSalary: 826
AvgSalary: 5160.184782608696

_id: "Female"
MaxSalary: 9925
MinSalary: 509
AvgSalary: 4954.861111111111

Task 2

1) A) Repeat the same processes but search for “Master” in education.

\$match [Open docs](#)

```

1 {
2   education: "Master",
3   age: {$gte: 21}
4 }

```

STAGE OUTPUT
Sample of 10 documents

_id: ObjectId('6ac79e9af0edfa62527a8c1')
firstName: "Evelyn"
lastName: "Wells"
gender: "female"
age: 24
email: "e.wells@randatamail.com"
education: "Master"
salary: 2923
maritalstatus: "Single"

_id: ObjectId('6ac79e9af0edfa62527a8c2')
firstName: "Martin"
lastName: "Alexander"
gender: "Male"
age: 26
email: "m.alexander@randatamail.com"
education: "Master"
salary: 2739
maritalstatus: "Single"

_id: ObjectId('6ac79e9af0edfa62527a8de')
firstName: "Paul"
lastName: "Johnston"
gender: "Male"
age: 25
email: "p.johnston@randatamail.com"
education: "Master"
salary: 2693
maritalstatus: "Single"

_id: ObjectId('6ac79e9af0edfa62527a8f1')
firstName: "Patge"
lastName: "Harper"
gender: "Female"
age: 38
email: "p.harper@randatamail.com"
education: "Master"
salary: 4021
maritalstatus: "Single"

_id: ObjectId('6ac79e9af0edfa62527a8f2')
firstName: "Miller"
lastName: "Foster"
gender: "Male"
age: 27
email: "m.foster@randatamail.com"
education: "Master"
salary: 4432
maritalstatus: "Single"

_id: ObjectId('6ac79e9af0edfa62527a8f7')
firstName: "Audrey"
lastName: "Sullivan"
gender: "Female"
age: 39
email: "a.sullivan@randatamail.com"
education: "Master"
salary: 5679
maritalstatus: "Married"

B) Find the avg, min, max salary grouped by marital status

Stage 1 ☒

```

1 {
2   _id: "$maritalstatus",
3   AvgSalary: {$avg: "$salary"},
4   MinSalary: {$min: "$salary"},
5   MaxSalary: {$max: "$salary"}
6 }

```

Output preview after [\\$group](#) stage (Sample of 2 documents)

_id: "Single"
AvgSalary: 5026.617647058823
MinSalary: 509
MaxSalary: 9989

_id: "Married"
AvgSalary: 5072.928571428572
MinSalary: 516
MaxSalary: 9989

2) Find min, max and avg salary of each age group of female

Stage 1:

<

Stage 1: \$match

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STAGE INPUT

Sample of 10 documents

email: "g.nelson@randatmai...

education: "Bachelor"

salary: 5347

maritalstatus: "Single"

_id: ObjectId('6ac79e9eaf0...

firstName: "Justin"

lastName: "West"

gender: "Male"

age: 27

email: "j.west@randatmai...

education: "Doctoral"

salary: 5783

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Daryl"

lastName: "Johnson"

gender: "Male"

age: 20

email: "d.johnson@randatma...

education: "Upper secondar...

salary: 4450

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Tiana"

lastName: "Fowler"

gender: "Female"

age: 27

email: "t.fowler@randatmai...

education: "Primary"

salary: 3529

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Alen"

lastName: "Barnes"

gender: "Male"

age: 26

email: "a.barnes@randatmai...

education: "Upper secondar...

salary: 6332

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Kirsten"

lastName: "Allen"

gender: "Female"

age: 21

email: "k.allen@randatmai...

education: "Lower secondar...

salary: 5792

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Charlie"

lastName: "Perkins"

gender: "Male"

\$match

Open docs

1 /**

2 * query: The query in MQL.

3 */

4 {

5 "gender" : "Female"

6 }

STAGE OUTPUT

Sample of 10 documents

_id: ObjectId('6ac79e9eaf0...

firstName: "Grace"

lastName: "Nelson"

gender: "Female"

age: 21

email: "g.nelson@randatmai...

education: "Bachelor"

salary: 5347

maritalstatus: "Single"

_id: ObjectId('6ac79e9eaf0...

firstName: "Tiana"

lastName: "Fowler"

gender: "Female"

age: 27

email: "t.fowler@randatmai...

education: "Primary"

salary: 3529

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Kirsten"

lastName: "Allen"

gender: "Female"

age: 21

email: "k.allen@randatmai...

education: "Lower secondar...

salary: 5792

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Florrie"

lastName: "Reed"

gender: "Female"

age: 19

email: "f.reed@randatmai...

education: "Upper secondar...

salary: 5497

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Amber"

lastName: "Brooks"

gender: "Female"

age: 27

email: "a.brooks@randatmai...

education: "Lower secondar...

salary: 1684

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Alberta"

lastName: "Robinson"

gender: "Female"

age: 27

email: "a.robinson@randatm...

education: "Lower secondar...

salary: 9319

maritalstatus: "Single"

Stage 2:

Stage 2: \$group

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STAGE INPUT

Options

\$group

Open docs

STAGE OUTPUT

Options

Sample of 10 documents

email: "g.nelson@randatmai...

education: "Bachelor"

salary: 5347

maritalstatus: "Single"

_id: ObjectId('6ac79e9eaf0...

firstName: "Tiana"

lastName: "Fowler"

gender: "Female"

age: 27

email: "t.fowler@randatmai...

education: "Primary"

salary: 3529

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Kirsten"

lastName: "Allen"

gender: "Female"

age: 21

email: "k.allen@randatmai...

education: "Lower secondar...

salary: 5792

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Florrie"

lastName: "Reed"

gender: "Female"

age: 19

email: "f.reed@randatmai...

education: "Upper secondar...

salary: 5497

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Amber"

lastName: "Brooks"

gender: "Female"

age: 27

email: "a.brooks@randatmai...

education: "Lower secondar...

salary: 1684

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...

firstName: "Alberta"

lastName: "Robinson"

gender: "Female"

age: 27

email: "a.robinson@randatm...

education: "Lower secondar...

salary: 9319

maritalstatus: "Single"

_id: ObjectId('6ac79e9eaf0...

firstName: "Rebecca"

lastName: "Douglas"

1 {

2 _id: "\$age",

3 AvgSalary: {\$avg:"\$salary"},

4 MinSalary: {\$min:"\$salary"},

5 MaxSalary: {\$max:"\$salary"}

6 }

7

Sample of 10 documents

_id: 22

AvgSalary: 5782

MinSalary: 3565

MaxSalary: 9925

_id: 20

AvgSalary: 5154

MinSalary: 1786

MaxSalary: 8539

_id: 19

AvgSalary: 4330.909090909091

MinSalary: 516

MaxSalary: 9846

_id: 21

AvgSalary: 3576.777777777778

MinSalary: 901

MaxSalary: 5792

_id: 27

AvgSalary: 4720

MinSalary: 1028

MaxSalary: 9319

_id: 24

AvgSalary: 4329

MinSalary: 2078

MaxSalary: 6921

_id: 28

AvgSalary: 5491

MinSalary: 646

MaxSalary: 9219

_id: 26

AvgSalary: 5319.5

MinSalary: 509

MaxSalary: 9611

_id: 25

AvgSalary: 5401

MinSalary: 707

MaxSalary: 9771

_id: 29

AvgSalary: 6169.222222222223

MinSalary: 2030

MaxSalary: 9913

3) Find min, max and avg salary of each age group of male
Stage 1:

Stage 1: \$match

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ADD STAGE

STAGE INPUT

Sample of 10 documents

email: "g.nelson@randatmai...
education: "Bachelor"
salary: 5347
maritalstatus: "Single"

_id: ObjectId('6ac79e9eaf0...
firstName: "Justin"
lastName: "West"
gender: "Male"
age: 27
email: "j.west@randatmail...
education: "Doctoral"
salary: 5783
maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...
firstName: "Daryl"
lastName: "Johnson"
gender: "Male"
age: 20
email: "d.johnson@randatma...
education: "Upper secondar...
salary: 4450
maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...
firstName: "Tiana"
lastName: "Fowler"
gender: "Female"
age: 27
email: "t.fowler@randatmai...
education: "Primary"
salary: 3529
maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...
firstName: "Alen"
lastName: "Barnes"
gender: "Male"
age: 26
email: "a.barnes@randatmai...
education: "Upper secondar...
salary: 6332
maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...
firstName: "Kirsten"
lastName: "Allen"
gender: "Female"
age: 21
email: "k.allen@randatmail...
education: "Lower secondar...
salary: 5792
maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...
firstName: "Charlie"
lastName: "Perkins"
gender: "Male"

\$match

Open docs

1 /**
2 * query: The query in MQL.
3 */
4 {
5 "gender" : "Male"
6 }

STAGE OUTPUT

Sample of 10 documents

_id: ObjectId('6ac79e9eaf0...
firstName: "Justin"
lastName: "West"
gender: "Male"
age: 27
email: "j.west@randatmail...
education: "Doctoral"
salary: 5783
maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...
firstName: "Daryl"
lastName: "Johnson"
gender: "Male"
age: 20
email: "d.johnson@randatma...
education: "Upper secondar...
salary: 4450
maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...
firstName: "Alen"
lastName: "Barnes"
gender: "Male"
age: 26
email: "a.barnes@randatmai...
education: "Upper secondar...
salary: 6332
maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...
firstName: "Charlie"
lastName: "Perkins"
gender: "Male"
age: 28
email: "c.perkins@randatma...
education: "Bachelor"
salary: 3586
maritalstatus: "Single"

_id: ObjectId('6ac79e9eaf0...
firstName: "Bruce"
lastName: "Cooper"
gender: "Male"
age: 22
email: "b.cooper@randatmai...
education: "Upper secondar...
salary: 5183
maritalstatus: "Single"

_id: ObjectId('6ac79e9eaf0...
firstName: "Dale"
lastName: "Payne"
gender: "Male"
age: 28
email: "d.payne@randatmail...
education: "Primary"
salary: 1065
maritalstatus: "Married"

Stage 2:

Stage 2: \$group

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ADD STAGE

STAGE INPUT

OPTIONS

Sample of 10 documents

email: "j.west@randatmail..."

education: "Doctoral"

salary: 5783

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...')

firstName: "Daryl"

lastName: "Johnson"

gender: "Male"

age: 20

email: "d.johnson@randatma..."

education: "Upper secondar..."

salary: 4450

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...')

firstName: "Alen"

lastName: "Barnes"

gender: "Male"

age: 26

email: "a.barnes@randatmai..."

education: "Upper secondar..."

salary: 6332

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...')

firstName: "Charlie"

lastName: "Perkins"

gender: "Male"

age: 28

email: "c.perkins@randatma..."

education: "Bachelor"

salary: 3586

maritalstatus: "Single"

_id: ObjectId('6ac79e9eaf0...')

firstName: "Bruce"

lastName: "Cooper"

gender: "Male"

age: 22

email: "b.cooper@randatmai..."

education: "Upper secondar..."

salary: 5183

maritalstatus: "Single"

_id: ObjectId('6ac79e9eaf0...')

firstName: "Dale"

lastName: "Payne"

gender: "Male"

age: 28

email: "d.payne@randatmail..."

education: "Primary"

salary: 1065

maritalstatus: "Married"

_id: ObjectId('6ac79e9eaf0...')

firstName: "Carl"

lastName: "Morrison"

\$group

Open docs

1 {

2 _id: "\$age",

3 AvgSalary: {\$avg:"\$salary"},

4 MinSalary: {\$min:"\$salary"},

5 MaxSalary: {\$max:"\$salary"}

6 }

7

STAGE OUTPUT

OPTIONS

Sample of 10 documents

_id: 19

AvgSalary: 5469.75

MinSalary: 1221

MaxSalary: 9543

_id: 22

AvgSalary: 4458.285714285715

MinSalary: 2158

MaxSalary: 8430

_id: 20

AvgSalary: 5309.333333333333

MinSalary: 1258

MaxSalary: 9587

_id: 27

AvgSalary: 5337.875

MinSalary: 1432

MaxSalary: 7548

_id: 26

AvgSalary: 5729.5

MinSalary: 826

MaxSalary: 8380

_id: 30

AvgSalary: 5363.090909090909

MinSalary: 1260

MaxSalary: 9989

_id: 29

AvgSalary: 7562.5

MinSalary: 5226

MaxSalary: 9899

_id: 28

AvgSalary: 5649.888888888889

MinSalary: 836

MaxSalary: 9989

_id: 25

AvgSalary: 4326.5

MinSalary: 2032

MaxSalary: 7667

_id: 21

AvgSalary: 4426.25

MinSalary: 1810

MaxSalary: 9460

4) Count married and unmarried females and males

\$group

Open docs

```
1 {
2   _id: {
3     maritalstatus : "$maritalstatus",
4     gender : "$gender"},
5   count: {$sum: 1}
6 }
```

STAGE OUTPUT

OPTIONS

Sample of 4 documents

▼ _id: Object (2)

maritalstatus: "Married"

gender: "Female"

count: 48

▼ _id: Object (2)

maritalstatus: "Single"

gender: "Female"

count: 60

▼ _id: Object (2)

maritalstatus: "Married"

gender: "Male"

count: 50

☑ _id: Object (2)

maritalstatus: "Single"

gender: "Male"

count: 42